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4. Build an Artificial Neural Network by implementing the Backpropagation algorithm and test the same using appropriate data sets.

1.

Study Hours	Attendance	Pass
8	90	Yes
6	85	Yes
2	60	No
3	65	No
7	88	Yes

PROGRAM

```
import numpy as np
X = np.array([[8,90],[6,85],[2,60],[3,65],[7,88]]) / [10,100]
y = np.array([[1],[1],[0],[0],[1]])
w = np.random.rand(2,1)
b = 0.1
for _ in range(10000):
    z = X @ w + b
    p = 1/(1+np.exp(-z))
    e = y-p
    w += 0.1 * X.T @ e
    b += 0.1 * e.sum()
sample = np.array([[7,85]]) / [10,100]
pred = 1/(1+np.exp(-(sample @ w+b)))
print("Prediction:", "Pass" if pred[0,0] > .5 else "Fail")
```

OUTPUT

Prediction: Pass

2.

Income	Credit Score	Approved
8	780	Yes
7	750	Yes
3	580	No
4	620	No
9	800	Yes

PROGRAM

```
import numpy as np
```

```
X = np.array([[8,780],[7,750],[3,580],[4,620],[9,800]]) / [10,800]
```

```
y = np.array([[1],[1],[0],[0],[1]])
```

```
w = np.random.rand(2,1)
```

```
b = 0.1
```

```
for _ in range(10000):
```

```
    p = 1/(1+np.exp(-(X@w+b)))
```

```
    e = y-p
```

```
    w += .1 * X.T @ e
```

```
    b += .1 * e.sum()
```

```
sample = np.array([[8,760]]) / [10,800]
```

```
p = 1/(1+np.exp(-(sample@w+b)))
```

```
print("Loan:", "Approved" if p[0,0]>.5 else "Rejected")
```

OUTPUT

Loan: Approved

3.

Temperature	Heart Rate	Disease
39	110	Yes
38.5	108	Yes
37	78	No
36.8	75	No
39.2	115	Yes

PROGRAM

```
import numpy as np
X = np.array([[39,110],[38.5,108],[37,78],
              [36.8,75],[39.2,115]]) / [40,120]
y = np.array([[1],[1],[0],[0],[1]])
w = np.random.rand(2,1)
b = 0.1
for _ in range(10000):
    p = 1/(1+np.exp(-(X@w+b)))
    e = y-p
    w += .1 * X.T @ e
    b += .1 * e.sum()
sample = np.array([[39,112]]) / [40,120]
p = 1/(1+np.exp(-(sample@w+b)))
print("Disease:", "Positive" if p[0,0]>.5 else "Negative")
```

OUTPUT

Disease: Positive

4.

Area	Bedrooms	Price
1200	2	Low
1500	3	Medium
2000	4	High
1000	2	Low
2200	4	High

PROGRAM

```
import numpy as np
X = np.array([[1200,2],[1500,3],[2000,4],
              [1000,2],[2200,4]]) / [2500,5]
y = np.array([[0],[1],[1],[0],[1]])
w = np.random.rand(2,1)
b = 0.1
for _ in range(10000):
    p = 1/(1+np.exp(-(X@w+b)))
    e = y-p
    w += .1 * X.T @ e
    b += .1 * e.sum()
sample = np.array([[2100,4]]) / [2500,5]
p = 1/(1+np.exp(-(sample@w+b)))
print("House Price:", "High" if p[0,0]>.5 else "Low")
```

OUTPUT

House Price: High

5.

Weight	Sweetness	Apple
150	90	Yes
160	88	Yes
145	92	Yes
120	70	No
125	72	No

PROGRAM

```
import numpy as np
X = np.array([[150,90],[160,88],[145,92],
              [120,70],[125,72]]) / [200,100]
y = np.array([[1],[1],[1],[0],[0]])
w = np.random.rand(2,1)
b = 0.1
for _ in range(10000):
    p = 1/(1+np.exp(-(X@w+b)))
    e = y-p
    w += .1 * X.T @ e
    b += .1 * e.sum()
sample = np.array([[155,90]]) / [200,100]
p = 1/(1+np.exp(-(sample@w+b)))
print("Fruit:", "Apple" if p[0,0]>.5 else "Not Apple")
```

OUTPUT

Fruit: Apple